# Problem Set 2

Due Online: Thursday, February 20, 11:59 pm

1. Classify the following models as $ARMA(p,q)$ (i.e., find the values of $p$ and $q$. Be careful about parameter redundancy), and determine whether they are stationary and invertible.

   (a) $y_t = y_{t-1} + \varepsilon_t - \frac{1}{2}\varepsilon_{t-1} - \frac{1}{2}\varepsilon_{t-2}$,

   (b) $y_t = 2y_{t-1} - 2y_{t-2} + \varepsilon_t - \frac{3}{4}\varepsilon_{t-1}$,

   (c) $y_t = \frac{5}{4}y_{t-1} + \frac{3}{2}y_{t-2} + \varepsilon_t - 2\varepsilon_{t-1}$,

   where $\varepsilon_t \sim WN(0,1)$ in all processes.

2. Consider the following $ARMA(2,1)$ model:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

   Show that the Yule-Walker equations are given by

$$\begin{aligned}
\gamma_0 &= \phi_1\gamma_1 + \phi_2\gamma_2 + \left(\theta_1\phi_1 + \theta_1^2 + 1\right)\sigma^2, \\
\gamma_1 &= \phi_1\gamma_0 + \phi_2\gamma_1 + \theta_1\sigma^2, \\
\gamma_h &= \phi_1\gamma_{h-1} + \phi_2\gamma_{h-2}, \quad \text{for } h \geq 2.
\end{aligned}$$

3. Consider the following $AR(2)$ process:

$$y_t = 3 + 0.3y_{t-1} + 0.04y_{t-2} + \varepsilon_t,$$

   where $\{\varepsilon_t\}$ is a Gaussian white noise process with variance 0.25. Suppose that $y_{99} = 4$ and $y_{100} = 3.5$.

   (a) Compute the 1-step- and 2-step-ahead point forecasts and their standard errors at the forecast origin $t = 100$. Construct 95% prediction intervals for the forecasts.

   (b) Compute the 100-step-ahead point forecast and its standard error at the forecast origin $t = 100$.

4. Consider the following stationary and invertible $ARMA(1,1)$ model:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1},$$

   for $t = 1, 2, \ldots, T$, where $\varepsilon_t \sim WN\left(0, \sigma^2\right)$. Suppose that the forecast origin is at time $t = T$.

(a) Find the 1-step-ahead forecast, given $y_T$ and $\varepsilon_T$. (*Comment*: Recall that $\varepsilon_T$ can be estimated using realizations $y_T, y_{T-1}, \ldots$, and hence we can consider it known at time $T$.)

(b) Show that starting from $h = 2$, the $h$-step-ahead forecast decays exponentially at the rate $\phi_1$ toward the unconditional mean of $\{y_t\}$, as $h$ increases.

(c) Show that the variance of the prediction error grows as we forecast further into the future.

5. (Computer Exercise) Consider the US quarterly real GDP growth from 1947Q2 to 2019Q4. The data can be retrieved using the following code in R, or from the BEA website: `https://fred.stlouisfed.org/series/A191RL1Q225SBEA`.

```
library(quantmod)
gdp.full <- -getSymbols("A191RL1Q225SBEA", src = "FRED", auto.assign = FALSE)
gdp <- -gdp.full["/2019"]
```

where the last line gets all the data until the end of 2019.

(a) Examine the time and ACF plots of the real GDP growth rates, *gdp*. Does the series look like stationary?

(b) Estimate three candidate models, AR(1), MA(1), and ARMA(1,1), for *gdp*. For each model, write down the estimated equation and investigate whether the model is adequate by examining the time plot, ACF, and the Ljung-Box test statistic of the residuals. *Comment*: Note that in the `arima()` output, the estimated mean is labeled as "intercept".

(c) Find the best *AR* model for *gdp*, including diagnostic checking. Write down the fitted model you end up choosing.

*Comment*: I am giving you more flexibility in making modeling decisions as we progress. You should consider all the tools we have learned so far in selecting the best model, including the ACF, PACF, AIC, BIC, the `auto.arima()` function, and diagnostic checking. To simplify the exercise, consider the maximum lag $p_{max} = 4$ and ignore seasonality. When using `auto.arima()`, please configure the `seasonal`, `max.p` and `max.q` arguments appropriately.

Answer the following questions using the model you selected in part (c).

(d) Does the model imply the existence of business cycles? Explain. Compute the average length of the cycles if they are present. (You can use the `polyroot()` function in R to find zeros of a polynomial.)

(e) Predict the $h$-step-ahead GDP growth rates at the forecast origin 2019Q4, for $h = 1, 2, 3, 4$. Find the 95% prediction intervals for these point forecasts.

(f) Predict the $h$-step-ahead GDP growth rates at the forecast origin 2018Q4, for $h = 1, 2, 3, 4$. Find the 95% prediction intervals for these point forecasts. Compare the forecasts with realized values of GDP growths.